PORTFOLIO · 2026

Analytical Portfolio

E-Commerce Conversion Intelligence & Behavioral Funnel Optimization

Three production-grade analytical pipelines spanning 2.75M e-commerce events, events, $53.5M in tracked financial transactions, and 100K+ retail orders — built with SQL, Python, PostgreSQL, Power BI, and DAX to ship measurable business impact.

SQL · PYTHON · POSTGRESQL · POWER BI · DAX · EXCEL

- EVENTS **2.75M**
- REVENUE **$53.5M**
- ORDERS **100K+**
- DELINQUENCY **6.06%**
- CHURN **85%**

PORTFOLIO OVERVIEW

Three projects. One pipeline philosophy.

End-to-end analytical work spanning behavioral analytics, financial data warehousing, and customer retention — each shipping a concrete, measurable business impact.

01

SQL · PYTHON · POWER BI

Behavioral Funnel from 2.75M Events

2.75M

RAW EVENTS PROCESSED

- 30-min inactivity session rule via LAG() window window functions
- 5 A/B tests with t-tests, Chi-Square, ANOVA (p 0.05)
- 4-quadrant 'Hidden Gem' framework = $50K untapped revenue

github.com/mojahid1252/Retailrocket-Ecommerce-Business-Analysis

02

POSTGRESQL · POWER BI · DAX

Star Schema Tracking $53.5M Revenue

$53.5M

TOTAL REVENUE TRACKED

- 52 weeks of transactions, 10,000+ records in star schema
- 20+ DAX measures with SUMX iterators + CALCULATE filters
- 2-page executive dashboard, 39 visuals, 15+ hrs/wk saved

github.com/mojahid1252/Credit-Card-Financial-Customer-Analytics-Dashboard

03

PYTHON · SQL · POWER BI · EXCEL

11-Tier RFM Segmentation of 96K 96K Buyers

85%

MONTH-1 CHURN RATE EXPOSED

- 100K+ orders EDA exposing 97% one-time buyer rate
- CLV + Pareto: 20% categories = 80% revenue
- $120K revenue opportunity via At-Risk re-engagement

github.com/mojahid1252/E-Commerce-olist-Business-Performance-Analysis

PROJECT 01 / 03

Behavioral Funnel from 2.75M Raw Events

SQL · Python · Power BI

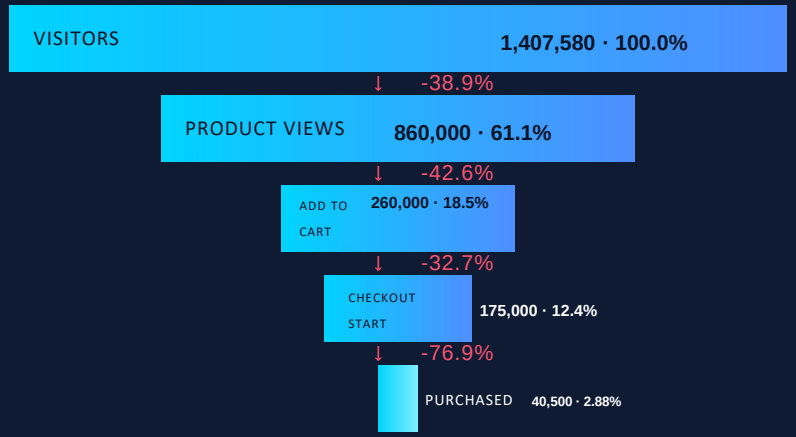
Engineered an end-to-end analytical pipeline processing 2.75 million raw e-e-commerce events. Implemented a 30-minute inactivity rule using LAG() window functions to accurately define user sessions and browsing behavior — behavior — surfacing a 76.8% cart abandonment profile and quantifying stage of funnel drop-off.

EVENTS PROCESSED	2.75M
CART ABANDONMENT	76.8%
POWER BI PAGES	8

github.com/mojahid1252/Retailrocket-Ecommerce-Business-Analysis

CONVERSION FUNNEL

76.8% CART ABANDONMENT



8-page interactive Power BI dashboard · cohort retention matrices · matrices · funnel drop-offs

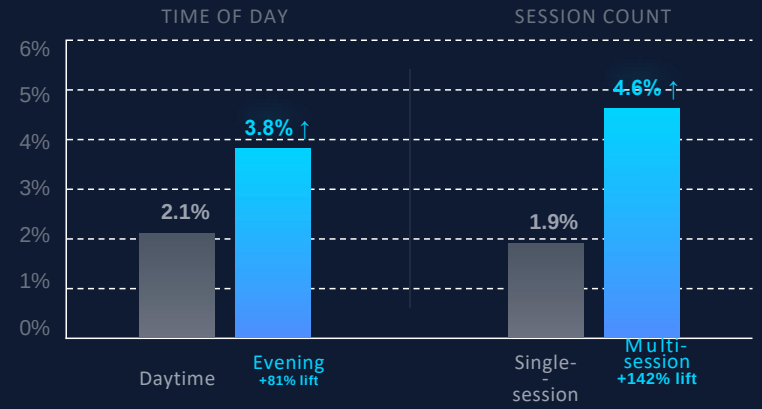
STATISTICAL HYPOTHESIS TESTING

A/B Tests Surface $50K Hidden Revenue

5 A/B tests evaluated using t-tests, Chi-Square, and ANOVA in Python. Evening users and multi-session users convert at significantly higher rates ($p < 0.05$) — informing targeted marketing and uncovering untapped revenue.

CONVERSION RATE BY SEGMENT

P < 0.05



T-TEST	CHI-SQ	ANOVA
5 tests	applied	applied

Identified evening + multi-session users as high-converting segments for targeted targeted marketing.

PRODUCT CLASSIFICATION FRAMEWORK

4-QUADRANT MATRIX

CONVERSION RATE ↑

HIDDEN GEM
High conversion, zero traffic

$50K untapped revenue

WINNERS
Scale and amplify

QUESTION MARK
Investigate or retire

UNDERPERFORMER
Optimize or deprecate

TRAFFIC →

$50,000 UNTAPPED REVENUE POTENTIAL
No additional ad spend — pure segment targeting

PROJECT 02 / 03

Star-Schema Warehouse Tracking $53.5M Revenue

PostgreSQL · Power BI · DAX

Architected a centralized PostgreSQL data warehouse consolidating 52 weeks of transactional data into a normalized star schema — powering a 2-page executive dashboard with 39 visuals for proactive risk management.

52W · LIVE

$53.5M

TOTAL REVENUE TRACKED
52 weeks · live

RISK

6.06%

DELINQUENCY RATE
segment concentration

RECORDS

10,000+

TRANSACTIONAL RECORDS
normalized star schema

DAX

20+

DAX MEASURES BUILT
SUMX + CALCULATE

METHODOLOGY

Pipeline Architecture

01 STAR SCHEMA MODEL
Fact + dimension tables normalizing 52 weeks of transactions

02 DAX ITERATORS
SUMX + CALCULATE filters tracking revenue, delinquency, activation KPIs

03 INCREMENTAL BATCH LOAD
Week 53 load with SQL validation checks

04 DATA INTEGRITY
100% reconciliation between PostgreSQL and Power BI semantic model model

05 EXECUTIVE DASHBOARD
2 pages, 39 visuals, 39 risk-segment cuts

MANUAL REPORTING TIME SAVED

15+ hours / week

enabling proactive risk management strategies for premium card segment targeting

github.com/mojahid1252/Credit-Card-Financial-Customer-Analytics-Dashboard

PROJECT 03 / 03

11-Tier RFM Segmentation of 96K Customers

Python · SQL · Power BI · Excel

RFM CUSTOMER SEGMENTS

96,000+
CUSTOMERS



Champions	8%	Loyal	12%	Pot. Loyalist	9%	Recent	7%
Promising	6%	Needs Att.	10%	About Sleep	6%	At Risk	18%
Can't Lose	4%	Hibernating	8%	Lost	12%	Total	100%

CRITICAL FINDING

97% ONE-TIME BUYER RATE

Exposed via comprehensive EDA on 100K+ orders using Python (Pandas, Seaborn) and SQL.

CHAMPIONS
8% · High R+F+M · retain & reward

AT RISK
18% · win-back priority · $120K opportunity

LOST
12% · low reactivation ROI · ROI · suppress

REVENUE OPPORTUNITY

$120,000

Re-engagement of At-Risk customers + optimization of top-performing product categories.

github.com/mojahid1252/E-Commerce-olist-Business-Performance-Analysis

COHORT ANALYSIS · PARETO · CLV

Cohort Heatmap Exposes 85% Month-1 Churn

Cohort retention matrices + Pareto analysis + Customer Lifetime Value computation surfaced where revenue concentrates and where and where customers leak — driving a 3-page Power BI dashboard and Excel KPI summary.

COHORT RETENTION MATRIX

MONTH-1 CHURN: 85%

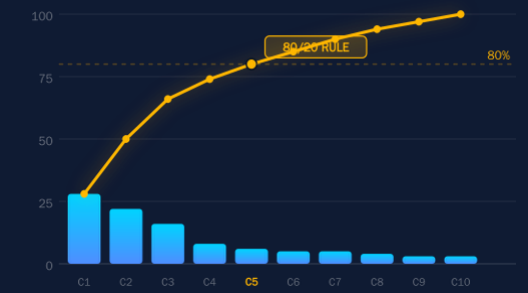
	M0	M1	M2	M3	M4	M5
JAN 2024	100%	15%	9%	6%	4%	3%
FEB 2024	100%	14%	8%	5%	3%	2%
MAR 2024	100%	16%	10%	6%	4%	3%
APR 2024	100%	13%	7%	5%	3%	2%
MAY 2024	100%	17%	11%	7%	5%	–
JUN 2024	100%	14%	8%	–	–	–

↓ M0→M1 drop averages 85% across cohorts

PARETO ANALYSIS

10 CATS

20% of categories drive 80% of revenue



Cat 1+2+3 = 66% revenue · top 5 = 80% revenue

3-PAGE POWER BI DASHBOARD

Geographic revenue distribution + late-late-delivery impact on review scores

EXCEL KPI SUMMARY SHEET

Tracks late delivery impact on customer customer review scores by region

CLV + PARETO + COHORTS

Customer Lifetime Value computed · cohorts cohorts exposed 85% M1 churn

THANK YOU

Let's build with data.

Three production-grade pipelines. Measurable business impact.

From 2.75M e-commerce events to a $53.5M financial warehouse to an 11-tier RFM segmentation model — every project here ships a concrete business outcome. Full source code, SQL, dashboards, and methodology available on GitHub.

01 · SQL · PYTHON · POWER BI

E-Commerce Conversion Intelligence & Behavioral Funnel Optimization

2.75M events · 5 A/B tests · $50K hidden revenue

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02 · POSTGRESQL · POWER BI · DAX

Financial Data Warehouse & Profitability Dashboard

$53.5M revenue · 6.06% delinquency · 15+ hrs/wk saved

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03 · PYTHON · SQL · POWER BI · EXCEL

E-Commerce Operations, Churn & Retention Analytics

100K+ orders · 11 RFM tiers · $120K revenue opportunity

github.com/mojahid1252/E-Commerce-olist-Business-Performance-Analysis

CORE STACK

SQL · Python · PostgreSQL · Power BI · DAX · Excel · Pandas · Seaborn · Statistical Testing